
PROJECT: Lake Kariba 500MW Floating Solar

Wind & Waves Study Report - Nyenje Bay, Lake Kariba

29 April 2026

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SYNOPSIS

This report presents a comprehensive wind, wave, and hydrodynamic assessment for the proposed 2 × 500 MW Floating Solar PV (FSPV) development at Nyenje (Gache-Gache) Bay, Lake Kariba, Zimbabwe, prepared primarily to support insurance due diligence and underwriting.

The study is based on a 50-year climatological assessment (1971–2020) using internationally recognised datasets and tools, notably ERA5 reanalysis, CCMP satellite winds, and the SWAN nearshore wave model. ERA5 wind data were conservatively bias-corrected (factor 0.718) using CCMP satellite data and validated against multiple local ground datasets, including Zimbabwe Meteorological Services (Kariba Airport), Padenga Holdings (13 years), and recent on-site weather station measurements.

Key findings indicate that Nyenje Bay experiences a low-energy and highly predictable wind and wave climate, dominated by diurnal lake–land breezes rather than storm-driven systems.

The values obtained are an order of magnitude lower than standard FSPV inland design limits (typically 1.0–1.5 m wave height), indicating minimal wave-induced structural loading.

Hydrological analysis demonstrates that even under extreme inflow or spillway discharge scenarios (up to 10,000 m³/s), resulting lake currents at the site are ≤0.5 m/s, posing no material risk to floating structures, moorings, or electrical systems.

The site benefits from strong natural protection, including, sheltered bay geometry, surrounding mountainous terrain, vegetated shorelines and limited effective fetch lengths

Based on long-term validated data, conservative modelling, and industry-standard methodologies, Nyenje Bay is confirmed as a technically suitable and insurable location for large-scale floating solar PV infrastructure. Environmental loading from wind, waves, and currents is very low, well within established engineering and insurance acceptance criteria, subject to standard design and operational controls.

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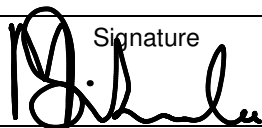
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PROJECT NAME: Lake Kariba 500MW Floating Solar

PROJECT DELIVERABLE: Wind & Waves Study Report - Nyenje Bay, Lake Kariba

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Table of Contents

1	Insurer Cover Note.....	6
1.1	Purpose of Cover Note	6
1.2	Study Basis and Reliability.....	6
1.3	Wind Climate – Insurability Summary.....	6
1.4	Wave Climate – Insurability Summary	7
1.5	Currents and Hydrology.....	7
1.6	Natural Risk Mitigation	7
1.7	Insurability Conclusion.....	7
2	Introduction	8
2.1	Executive Summary.....	8
2.2	Background & Objectives	10
2.2.1	Project Background.....	10
2.2.2	Project Objective.....	10
2.2.3	Study Area	10
3	Methodology & Execution.....	11
3.1	Data Acquisition & Process.....	12
3.2	ERA-5 Results for the Whole Lake (Data downloads; Processing; SWAN Extract) .	13
3.3	Data from Padenga Station for 13 years.....	16
3.4	Data from Lake harvest.....	16
3.5	Data from MSD - Kariba Airport. (2001 to 2005)	17
3.6	Wind Speed, Direction and Seasonal Patterns.....	17
3.7	Implications on Waves	17
3.8	Validation Process	17
3.9	Comparison of Ground Data	18
3.10	De-Biasing \ Correcting ERA-5 Data prior to input into SWAN Model.....	19
3.10.1	Dataset Characteristics:	19

	3.10.2 Results of De-Biasing	20
	3.10.3 Bias Factor Explanation.....	20
	3.10.4 Correlation of Results	21
	3.10.5 General Wind Statistics for Lake Kariba	22
4	SWAN Model Reporting	26
5	Hydrological Summary	30
6	Structural Design Criteria Memorandum	31
	6.1 Scope and Purpose.....	31
	6.2 Governing Codes and Standards.....	31
	6.3 Design Philosophy	32
	6.4 Site-Specific Environmental Design Parameters	32
	6.5 Classification of Actions	33
	6.6 Serviceability Limit State (SLS) Load Combinations	33
	6.6.1 SLS-1: Operational Condition	33
	6.6.2 SLS-2: Environmental Condition	34
	6.7 Ultimate Limit State (ULS) Load Combinations – EN / SANS.....	34
	6.7.1 ULS-W: Wind Governing.....	34
	6.7.2 ULS-WV: Wave Governing.....	34
	6.7.3 ULS-COMB: Combined Wind and Wave	35
	6.8 Ultimate Limit State – DNV Marine Combinations.....	35
	DNV-ULS	35
	6.9 Accidental / Extreme Limit State (ALS).....	35
	6.10 Mooring Design Load Combinations.....	36
	6.10.1 Mooring SLS	36
	6.10.2 Mooring ULS.....	36
	6.11 Design Acceptance Criteria	36
	6.11.1 Structural Members.....	36
	6.11.2 Floats	36

6.11.3	Connections	36
6.12	Summary	37
7	Conclusion	37

List of Tables

Table 2-1: Results (General to Lake: items 1 to 8 and specific to Site: items 9 to16)	8
Table 3-1: Tool Tab Functionality	11
Table 3-2: Data Set Characteristics (Vector Characterization of the Site)	12
Table 3-3: Data Set Characteristics (Vector Characterization of the Site)	18

List of Figures

Figure 2-1: Processing Tool Main Menu	9
Figure 3-1: ERA-5 Download over 50 Years	13
Figure 3-2: CCMP Download over 20 years.....	13
Figure 3-3: Onsite Weather Menu.....	14
Figure 3-4: Onsite Results (recently commissioned)	14
Figure 3-5: Onsite Wind Speed Averaged and Classes	15
Figure 3-6: Onsite Wind Turbulence & Direction	15
Figure 3-7: Onsite Wind Direction & Distribution	16
Figure 3-8: Processing ERA-5 Data.....	18
Figure 3-9: ERA-5 Data vs CCMP Bias Analysis.....	20
Figure 3-10: SWAN Extract.....	21
Figure 3-11: Wind Rose - North Easterly - consistent with data from MSD; Padenga & On site	22
Figure 3-12: Wind Stats in year of highest speed	22
Figure 3-13: Diurnal Results - show generally calm wind conditions over night over a 50-year period	23



Figure 3-14: Wind Averages & Gusts	23
Figure 3-15: Values for Wind speed averages & gusts	24
Figure 3-16: Annual Max Wind Speeds	24
Figure 3-17: JonSwap Typical Results.....	25
Figure 4-1: SWAN Home page.....	26
Figure 4-2: Common Bathymetric Information	26
Figure 4-3: Select Single Time Step.....	27
Figure 4-4: Selected Single Time Wave Height & Period Per Grid	27
Figure 4-5: Monthly Max Wave heights	28
Figure 4-6: Monthly Wave Height Stats	28
Figure 4-7: Yearly Max Wave Heights	29
Figure 4-8: Extreme Values And 50-100 Year Return Period.....	29
Figure 4-9: Return Periods - Wave Heights.....	30



List of Abbreviations and Acronyms

ECMWF	European Centre for Medium-Range Weather Forecasts
ERA-5	ECMWF Reanalysis 5th Generation
CCMP	Cross-Calibrated Multi-Platform
u ₁₀	Eastward wind component at 10m height
v ₁₀	Northward wind component at 10m height
GEBCO	General Bathymetric Chart of the Oceans
MSD	Meteorological Services Department
SWAN	Simulating WAVes Nearshore
H _s	Significant Wave Height
T _p	Peak Wave Period
FSPV	Floating Solar Photo Voltaic

1 Insurer Cover Note

Project:	Nyenje Bay Floating Solar PV Development (2 × 500 MW)
Location:	Nyenje (Gache-Gache) Bay, Lake Kariba, Zimbabwe
Document Basis:	Wind & Waves Study Report – Draft 4 (24 April 2026)
Prepared for:	Insurance Due Diligence and Underwriting Review

1.1 Purpose of Cover Note

This cover note summarizes the findings of the Wind & Waves Study undertaken for the proposed large-scale Floating Solar PV (FSPV) installation at Nyenje Bay, Lake Kariba, with specific focus on environmental loading relevant to insurability, including wind, waves, and lake currents.

1.2 Study Basis and Reliability

Assessment Period:	50-year historical climatology (1971–2020)
Primary Datasets:	ERA-5 reanalysis (ECMWF)
	CCMP satellite wind data
Model Used:	SWAN (Simulating Waves Nearshore), industry-standard spectral wave model
Validation:	Zimbabwe Meteorological Services (Kariba Airport)
	Padenga Holdings long-term wind records (~13 years)
	Recent on-site weather station data
Data Treatment:	ERA-5 wind speeds conservatively bias-corrected (factor 0.718) to avoid over-prediction of wind and wave loads

The methodology is consistent with internationally accepted engineering and insurance practice for inland water infrastructure.

1.3 Wind Climate – Insurability Summary

Mean Wind Speed:	2.4 m/s
Typical Maximum Wind Speeds:	Below 10 m/s
Observed Short-Duration Gusts:	14.8 m/s (on-site)
Prevailing Direction:	North-Easterly
50-Year Extreme Winds:	Well below common FSPV design and insurance thresholds (≥ 30 m/s)



Assessment: The site exhibits a low-energy, predictable wind regime, dominated by diurnal lake breezes rather than synoptic storm systems. Wind conditions present low structural and operational risk to floating solar platforms and associated mooring systems.

1.4 Wave Climate – Insurability Summary

Mean Significant Wave Height (H_s):	0.18 m
Maximum Modelled Wave Height:	0.21 m
50-Year Return Period H_s:	0.22 m
100-Year Return Period H_s:	0.23 m
Observed Local Conditions:	Generally, less than 0.30 m

The typical insured inland floating solar systems are designed for wave heights of 1.0–1.5 m. Nyenje Bay waves are an order of magnitude lower. Wave-induced loading is minimal and well within conservative design and insurance acceptance criteria.

1.5 Currents and Hydrology

Extreme Lake Flow (historical):	10,000 m ³ /s
Resulting Current at Site:	≤0.5 m/s (upper bound)
Typical Operating Currents:	Negligible

The currents at the site do not pose a material risk to floating structures, anchoring systems, or electrical infrastructure.

1.6 Natural Risk Mitigation

- Sheltered bay geometry
- Surrounding mountainous terrain
- Vegetated shorelines
- Limited fetch lengths

These features provide inherent protection, substantially reducing exposure to extreme wind-wave events.

1.7 Insurability Conclusion

Based on long-term validated data, conservative modelling, and internationally accepted analysis methods, the Nyenje Bay site demonstrates very low environmental loading risk for large-scale floating solar PV infrastructure.

It is concluded that the project is suitable for insurance coverage with respect to wind, wave, and hydrodynamic environmental risks, subject to standard engineering design compliance and routine construction and operational risk controls..



2 Introduction

2.1 Executive Summary

The 4th report largely corrects the derivation of the wind average & maximum speeds and gust factor over 3 minute and 10-minute periods as well as the respective gust over the 2 sec and 10 sec period. This is a wind and subsequent waves study for the proposed 2 x 500MW FSPV project at the Nyenje Bay (also known as Gache-Gache Bay) of Lake Kariba. The set of results in table 01 below is pleasing and technically confirms the suitability of the location for the proposed project. Data gathering was a combination of access to both the international databases (ERA-5 and CCMP) and local databases namely (MSD; Padenga Holdings; Lake Harvest and more recently own on-site weather station). Finally, to process that data into desired information, we developed a tool that can run various scenarios as will be demonstrated. A screen shot of the main menu of the tool is image 01 on next page and, this report shall go into details of process of deriving the different aspects of results.

Table 2-1: Results (General to Lake: items 1 to 8 and specific to Site: items 9 to16)

Item	Parameter Description	Determination	Comment
1	Mean wind speed	2.4 m/s	
2	Max wind speed	8.9m/s	
3	Prevailing wind direction	052 deg	
4	Hourly Mean	1.72; 6.39; 1	Mean or Avg (m/s); Max (m/s); Gust Factor
5	3 - Min Avg	1.72; 6.05; 1.15	
6	10 - Min Avg	1.72; 5.15; 1.00	
7	3 - Sec Gust	2.90; 9.27; 1.45	
8	10 - Sec Gust	3.05; 8.63; 1.35	
9	Mean wave height	0.176m	
10	Max wave height	0.206m	0.192 from JonSwap
11	Peak wave period	1.72 (s)	
12	Mean wave period	1.23 (s)	
13	Wave direction	North Easterly	
14	50 Yr Return period	0.217m	
15	100 yr Return period	0.225m	
16	FSPV Design Max Std for inland calm waters	Wind Speed 30 m/s & Wave Height 1.5m	Nyenje Bay is therefore an appropriate site for the proposed project
	FSPV Design Max Std for nearshore coastal	Wind Speed 30 m/s & Wave Height 2-3m	



	FSOV Design Max Std for offshore Marine	Speed ~ 30 m/s Height ~ >5m	
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This report shall be presented alongside our version 1.0 of the LINUX-based tool developed and used to derive the results. Various processing screen shots will be shared along with brief explanations.

Key Informatics Wind/Waves Analysis Tool - Nyenje Bay, Kariba

Complete wind, wave, and SWAN wave modeling analysis

 Data Acquisition & Process  General Wind Statistics  Generic Wave Analysis (JONSWAP)  SWAN Model  Solar Data

Data Acquisition & Processing

 ERA5 Download  CCMP Download  Process Data  SWAN Extract

Process ERA5 Data

Data Directory

./data/era5/raw

Target Latitude

-16.5300

- +

Target Longitude

28.8300

- +

 Found 600 files (600 NetCDF, 0 GRIB)

Figure 2-1: Processing Tool Main Menu

2.2 Background & Objectives

2.2.1 Project Background

Lake Kariba, one of the world's largest man-made lakes by volume, plays a crucial role in the region's hydroelectric power generation. Stretching a maximum length of 280km and width of 40km the resultant downstream industries of fisheries, tourism and transportation. are also very important economic activities to the region. The proposed FSPV project of 2 x 500MW at full development and the South Bank installed capacity of 1050MW will result in a hybridized power generation and evacuation scenario. Establishing the wind and wave climate of the lake at Nyenje bay is essential for determining the suitability of the bay as a FSPV site.

2.2.2 Project Objective

The primary objective of this report is to provide relevant technical information to enable a bankable determination and confirmation of the appropriateness of Nyenje site to host the 2 x 500MW FSPV project. To achieve these various activities that include but not limited to the following shall be executed:

- Establish a 50-year wind climatology for Lake Kariba using ERA-5 reanalysis data (1971-2020)
- Establish the satellite-based wind data CCMP (2001-2020) and apply for debiasing
- Validate wind fields against available ground observations (MSD; Padenga; Lake Harvest on-site)
- Run the SWAN wave model for Nyenje Bay using the Bathymetry results

2.2.3 Study Area

The study focuses on Lake Kariba, specifically the Nyenje Bay area located at approximately 16.53°S, 28.83°E. The domain extends from 26°E to 30°E longitude and 18°S to 10°S latitude, covering the entire lake and surrounding catchment.

3 Methodology & Execution

As mentioned in the Executive Summary, the developed tool became the basis of performing the scientific processing of raw data into engineering information as technical input. The tool has 4 key tabs, also expanded in the table below:

- Data acquisition & Process
- General Wind Statistics
- Generic \ Typical Wave Analysis (JonSwap)
- SWAN Model

Table 3-1: Tool Tab Functionality

Tab	Sub- Tabs	Comment
Data Acquisition & Process	A. Applied Datasets • ERA-5 Raw Dataset (1971-2020) • CCMP Dataset (2001-2020) • De-biased ERA-5 Data Set (50 years) B. Validation Ground Datasets: • MSD requested 50 years; received 5 years • Padenga got 13 years wind direction • Lake Harvest - still pending • Own on-site station - just 1 week old	
General Wind Statistics (Whole Lake)	Wind Rose (Direction) Statistics (mean speed; max speed; Diurnal) Wind Averages & Gusts • 10 min mean (x1.0) • min mean (x1.15) • 10 sec gusts (x1.35) • 3 sec gusts (x1.45)	• Ref wind speed • Dynamic response • Peak gust • Ultimate limit
Generic \ Typical Wave	JonSwap Workings	Typical results
SWAN Model (Site specific) Grid size: 9x8 points (72 pts) Domain: 4 km x 3.5 km Resolution: 500m Spectral Directions: 36 bins From Bathymetry info combined with de-biased or corrected ERA-5.	• Single time step Wave Analysis • Monthly Max Wave Height • Yearly Max Wave Height • Return periods (50-100 year)	

3.1 Data Acquisition & Process

Image 01 shows the available and applied two data-set downloads (ERA-5 and CCMP) and these two derive the ERA-5 corrected or de-biased version that is later used in the SWAN model. There are four other datasets to be housed under the Ground Validation namely (MSD data; Lake Harvest data; Padenga Holdings data and own on-site measurement data) as indicated in the first-row sub-tabs of table 3-1 above.

The ERA-5 accessed and used 50-year (1971-2020) data for the whole lake whilst the CCMP, which is satellite based, accessed and used the 20-year (2001-2020) data. The two sets of data resulted in the corrected or de-biasing the raw ERA-5 data by a 0.718 factor. It is the corrected ERA-5 data that would then be integrated with the determined Bathymetry results and processed into the SWAN model to yield more site-specific results. In addition, the actual site was further refined to 72 positions where the respective parameter derivations were performed.

Table 3-2: Data Set Characteristics (Vector Characterization of the Site)

Parameter	Details
Period	1971-2020 (50 years)
Temporal Resolution	Hourly
Spatial Resolution	0.25° × 0.25° (28km)
Domain	26°E-30°E, 18°S-10°S
Variables	u10, v10 (10m wind components)
	t2m (2m temperature)
File Format	NetCDF4, monthly files
Total Size	2.5 GB
Source	Copernicus Climate Change Service

The raw ERA5 wind components (u10, v10) are processed to derive wind speed and direction using:

$$\text{Wind Speed} = \sqrt{u_{10}^2 + v_{10}^2}$$

$$\text{Wind Direction} = 270 - \arctan2(v_{10}, u_{10}) \times \frac{180}{\pi} \bmod 360$$

where direction follows meteorological convention (0° = wind from North, clockwise).

3.2 ERA-5 Results for the Whole Lake (Data downloads; Processing; SWAN Extract)

 **Key Informatics Wind/Waves Analysis Tool - Nyenje Bay, Kariba**

Complete wind, wave, and SWAN wave modeling analysis

Key Concept: Bathymetry is CONSTANT (loaded once). Wind files are VARIABLE (one per hour from ERA5 debiased data).

Data Acquisition & Processing

ERA5 Download CCMP Download Process Data SWAN Extract

ERA5 Data Download

Download ERA5 reanalysis data from Copernicus Climate Data Store

Start Year: 1990 - + Region: Lake Kariba

End Year: 2020 - +

- Latitude: 18°S to 10°S
- Longitude: 26°E to 30°E

Download Options:

Figure 3-1: ERA-5 Download over 50 Years

 **Key Informatics Wind/Waves Analysis Tool - Nyenje Bay, Kariba**

Complete wind, wave, and SWAN wave modeling analysis

Key Concept: Bathymetry is CONSTANT (loaded once). Wind files are VARIABLE (one per hour from ERA5 debiased data).

Data Acquisition & Processing

ERA5 Download CCMP Download Process Data SWAN Extract

CCMP Satellite Data

CCMP wind data from NASA PODAAC

Download Instructions:

1. Visit: https://podaac.jpl.nasa.gov/dataset/CCMP_WINDS_10M6HR_L4_V3_1
2. Select area: 26°E to 30°E, 18°S to 10°S
3. Download NetCDF files

Figure 3-2: CCMP Download over 20 years

The original ERA-5 data and the CCMP data corrected the ERA-5 by a factor of 0.718 We expect to add ground validation datasets from four sources stated in table 3-1. This will be figures 3.3 to 3.7 below is nearby ground data progressively from vicinity to site to the airport.

Onsite Data from a recently commissioned Key Informatics Weather Station. One can track by visiting www.vitalweather.co.za. The login details are as follows: **USERNAME:** *keyinfo* and **PASSWORD:** *weather*.



ANALYTICS

Select a Station: Key Informatics 01 Kariba VIF206 ▼

Select Year: 2026 ▼

Show Graph:

- ☒ Wind Speed Averages
- ☒ Wind Rose
- ☒ Wind Speed Distribution Classes
- ☒ Wind Turbulence
- ☒ Wind Direction Distribution
- ☐ Temperature Averages
- ☐ Humidity Averages
- ☐ Rainfall
- ☐ Evapotranspiration
- ☐ Station Up-Time

Wind Speed scale: m/s ▼

Temperature scale: Celsius ▼

Rain scale: Millimeters ▼

Highs and Lows: ☒ Show data

ANALYSE

Figure 3-3: Onsite Weather Menu

Highest Wind Gust:	14.75	m/s	2026-04-14 13:45:00
Highest Recorded Average Wind Speed:	10.59	m/s	2026-04-14 13:45:00
Average Annual Wind Speed:	3.09	m/s	
Highest Recorded Temperature:	32.44	°C	2026-04-14 15:30:00
Lowest Recorded Temperature:	20.7	°C	2026-03-24 07:35:11
Average Annual Temperature:	24.7	°C	
Highest Recorded Humidity:	79.11	%	2026-04-14 06:30:12
Lowest Recorded Humidity:	48	%	2026-04-14 15:55:00
Average Annual Humidity:	67	%	
Highest Days Rain:	1	mm	2026-03-23 14:15:10
Total Annual Rain Fall:	1	mm	
Annual Station Up-Time:	2	%	

Figure 3-4: Onsite Results (recently commissioned)

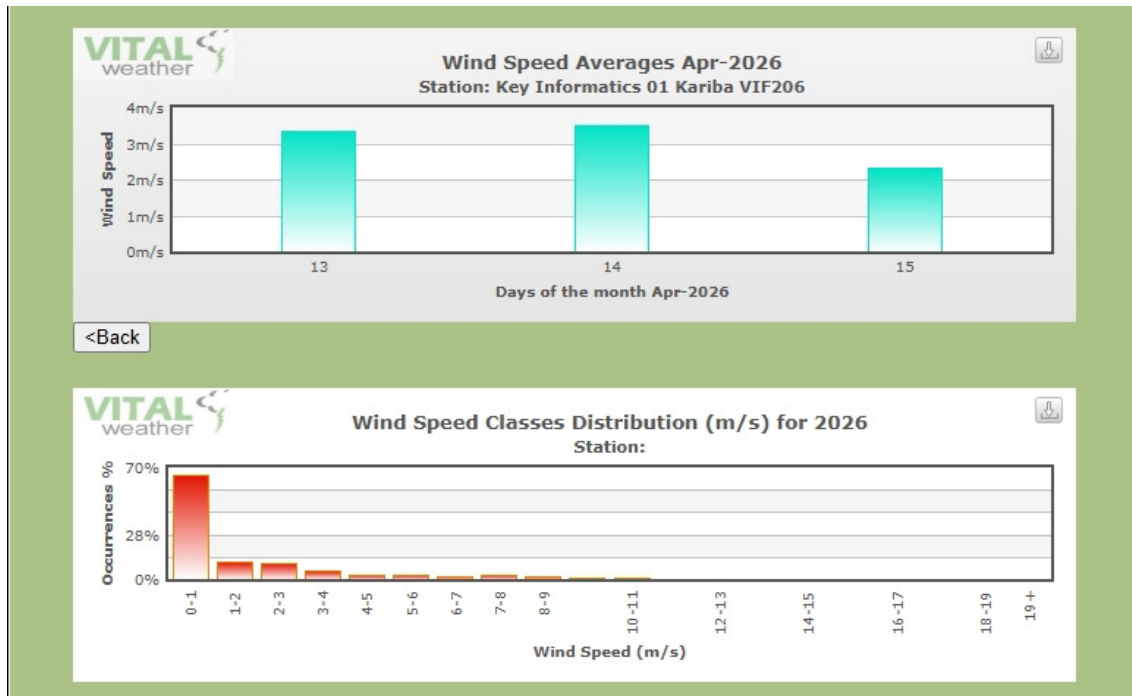


Figure 3-5: Onsite Wind Speed Averaged and Classes

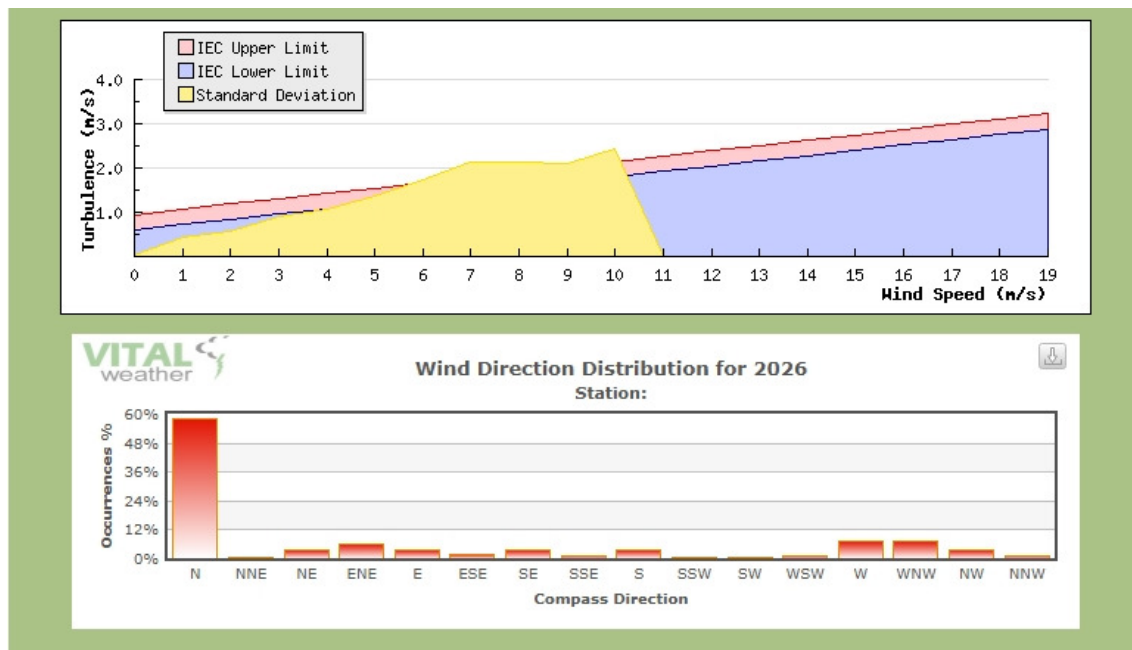


Figure 3-6: Onsite Wind Turbulence & Direction

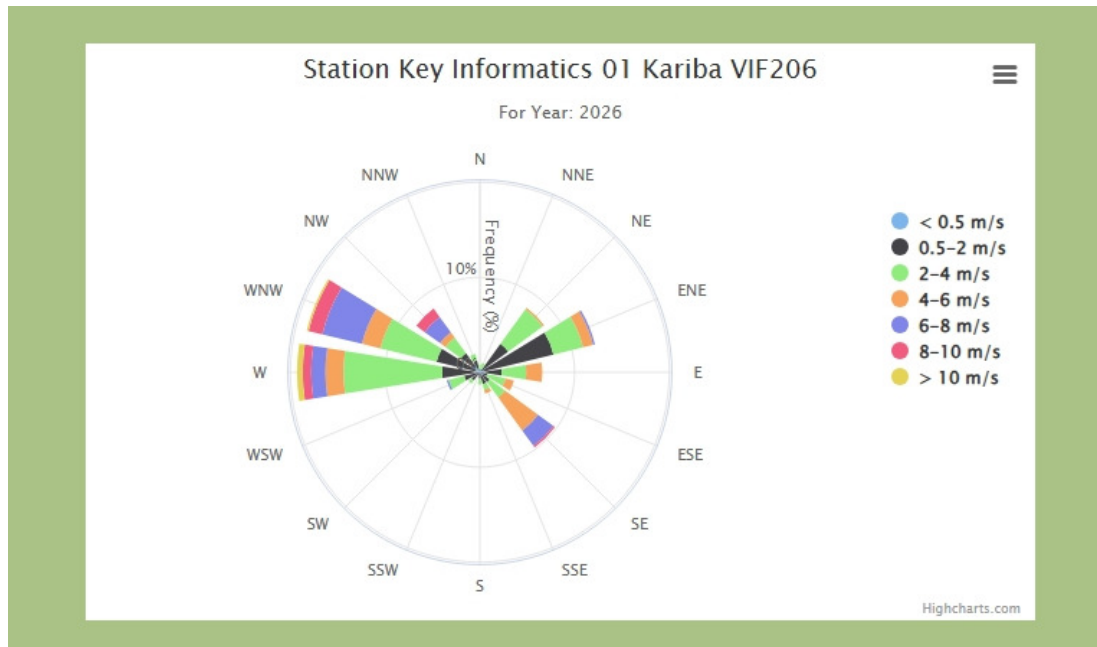


Figure 3-7: Onsite Wind Direction & Distribution

3.3 Data from Padenga Station for 13 years

Refer to Padenga_Weather_Station_Data_2013_2026.XLSM.

Wind Speed was not available and only direction was captured and overwhelmingly N (North) winds (by far the highest frequency) and Secondary contributions: NE / NNE / ENE (significant cluster) and SW / WNW / W (secondary but noticeable)

Seasonal Pattern from the monthly breakdown:

Wet season (Nov–Mar): Strong NE to ENE dominance

Dry season (May–Aug): Winds become more variable, with: Increased SW, WNW, W and still a persistent N component.

Transition months: Mixed but still north biased.

3.4 Data from Lake harvest

Data to follow.

3.5 Data from MSD - Kariba Airport. (2001 to 2005)

Refer to Kariba Wind Data 2001-2005.XLSM

3.6 Wind Speed, Direction and Seasonal Patterns

- **Average wind speed:** 2 knots (≈ 1.0 m/s) → generally *very light winds*
- **Median:** 1.6 knots → calm conditions dominate
- **Maximum observed:** 9.3 knots (≈ 4.8 m/s)
- **Typical range:** 1–3 knots
- **Most frequent direction code:** Winds are predominantly easterly to north-easterly
- **Based on standard 36-sector coding:** 70° (ENE – East-Northeast)

Monthly averages show a strong seasonal signal:

- **Strongest winds:**
 - October (~ 3.4 knots) → peak
 - September–November elevated
- **Caldest period:** February–March (~ 1.0 – 1.2 knots)
- Late dry season (Sep–Oct) = highest wind energy
- Rainy season (Jan–Mar) = very calm

3.7 Implications on Waves

Given both the low wind speeds and limited duration of strong winds, the lake has Calm to slight wave conditions. However, wave heights typically < 0.3 m most of the time and Occasionally 0.5 – 1.0 m during peak winds Lake Kariba is long (~ 280 km, east–west oriented) - the fetch becomes a key factor in the wave analysis and generation. Since prevailing winds are ENE (blowing toward WSW). Therefore, this creates firstly a long effective fetch along the lake axis and secondly moderate winds can generate noticeable waves over long distances. Practically however, one gets meaningful wave action when winds reach ~ 8 – 10 knots (4 to 5 m/s) and persist for several hours. It is under such conditions that waves can build to ~ 1 m or slightly more, especially in the central basin.

The Nyenje Bay site and orientation is advantageous in that it is not only neatly tucked in the eastern basin but also naturally shielded on 3 sides with mountains and well treed areas. There are very reduced chances of adverse waves though the SWAN model will do the specific run.

3.8 Validation Process

Metric	Value
Number of Matched Observations	1,826
Bias (Debiased ERA5 - Ground)	$+0.60 \text{ m s}^{-1}$
RMSE	0.95 m s^{-1}
Correlation Coefficient	0.40
Mean Absolute Direction Error	63

The modest correlation (0.40) is explained by:

- Spatial scale mismatch: 0.25 grid (approximately 28 km) vs point measurement
- Land-lake interface: Airport is on land, model needs lake winds
- Local topography: Terrain effects not captured by ERA5
- Temporal averaging: Hourly data vs instantaneous observations

3.9 Comparison of Ground Data

From the recent on-site data, we got these significant parameters (albeit just a fortnight) we found that Average wind speed: 3.09 m/s and Max gust: 14.75 m/s.

Table 3-3: Data Set Characteristics (Vector Characterization of the Site)

Parameter	On-Site	Padenga Site	Kariba Airport
Average Wind	3.1 m/s	Not given (assume similar to site)	3.5 to 5 m/s
Gusts	15 m/s	Not given (assume similar to site)	Slightly higher
Wind spread \ variability	N \ Medium	North \ Low	Bi direction \ High
Wind Turbulence	Mod-High	Moderate	High
General Comment	Practical	Similar to on-site & get Lake H	extreme envelope use

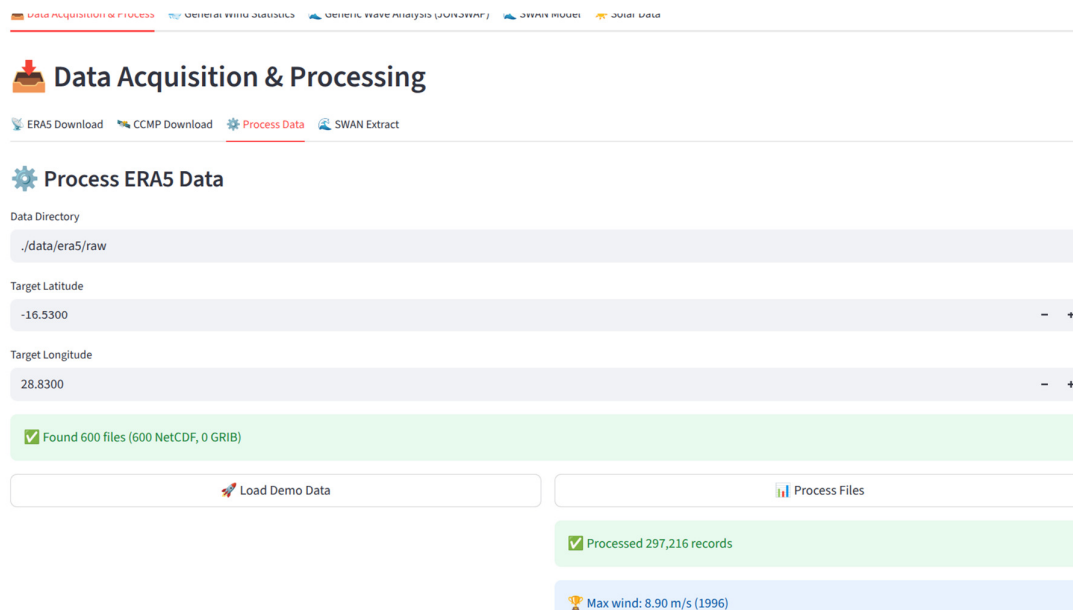


Figure 3-8: Processing ERA-5 Data

3.10 De-Biasing \ Correcting ERA-5 Data prior to input into SWAN Model

The model then considers wind bias analysis by comparing ERA-5 data with CCMP satellite observations over Lake Kariba for the same period 2001-2020. The analysis aims to quantify systematic differences between the two datasets and derive correction factors for improving ERA5 wind estimates.

ERA-5 is a state-of-the-art atmospheric reanalysis that combines model physics with observations. However, systematic biases can occur due to:

- Model parameterization limitations
- Sparse observation networks in data-sparse regions
- Topographic effects (Lake Kariba's complex shoreline)

CCMP (Cross-Calibrated Multi-Platform) satellite winds provide an independent, observation-based reference. By comparing these datasets, we can:

- Quantify ERA5 biases over Lake Kariba
- Develop statistical correction factors
- Improved wind inputs for SWAN wave modeling

Outputs received:

- Mean Wind Speed: ERA5 (2.60 m s^{-1}) is significantly higher than CCMP (1.87 m s^{-1})
- Bias: ERA5 overestimates wind speeds by $+0.74 \text{ m s}^{-1}$ on average
- Correlation: Strong agreement (0.89) between datasets
- Bias Factor: 0.718 – Multiply ERA5 by 0.718 to match CCMP

3.10.1 Dataset Characteristics:

Parameter	ERA5	CCMP
Period	2001-2020	2001-2020
Resolution	$0.28^\circ \times 0.28^\circ$, hourly	$0.25^\circ \times 0.25^\circ$, monthly
Variables	u10, v10	u, v
Source	ECMWF	NASA PO. DAAC

The process and analysis followed the following steps:

1. **Collocation:** ERA5 hourly data averaged to monthly means
2. **Sub setting:** Both datasets cropped to Lake Kariba region ($26\text{-}30^\circ\text{E}$, $18\text{-}10^\circ\text{S}$)
3. **Re-gridding:** CCMP interpolated to ERA5 grid
4. **Comparison:** Statistical analysis of paired wind speeds



3.10.2 Results of De-Biasing

Metric	Value	Interpretation
ERA5 Mean Wind Speed	2.60 m s ⁻¹	Reference (biased)
CCMP Mean Wind Speed	1.87 m s ⁻¹	"Truth" reference
Bias (ERA5 - CCMP)	+0.74 m s ⁻¹	ERA5 overestimates
RMSE	0.77 m s ⁻¹	Typical error magnitude
Correlation	0.89	Excellent pattern agreement
Bias Factor	0.718	Correction multiplier

The positive bias indicates that **ERA5 consistently overestimates wind speeds** across Lake Kariba compared to CCMP satellite observations. This means that raw ERA5 wind speeds would lead to overestimated wave heights in SWAN model simulations.

3.10.3 Bias Factor Explanation

$$ERA5_{corrected} = ERA5_{original} \times 0.718$$

The graphs below were derived from the model as the de-biasing process.

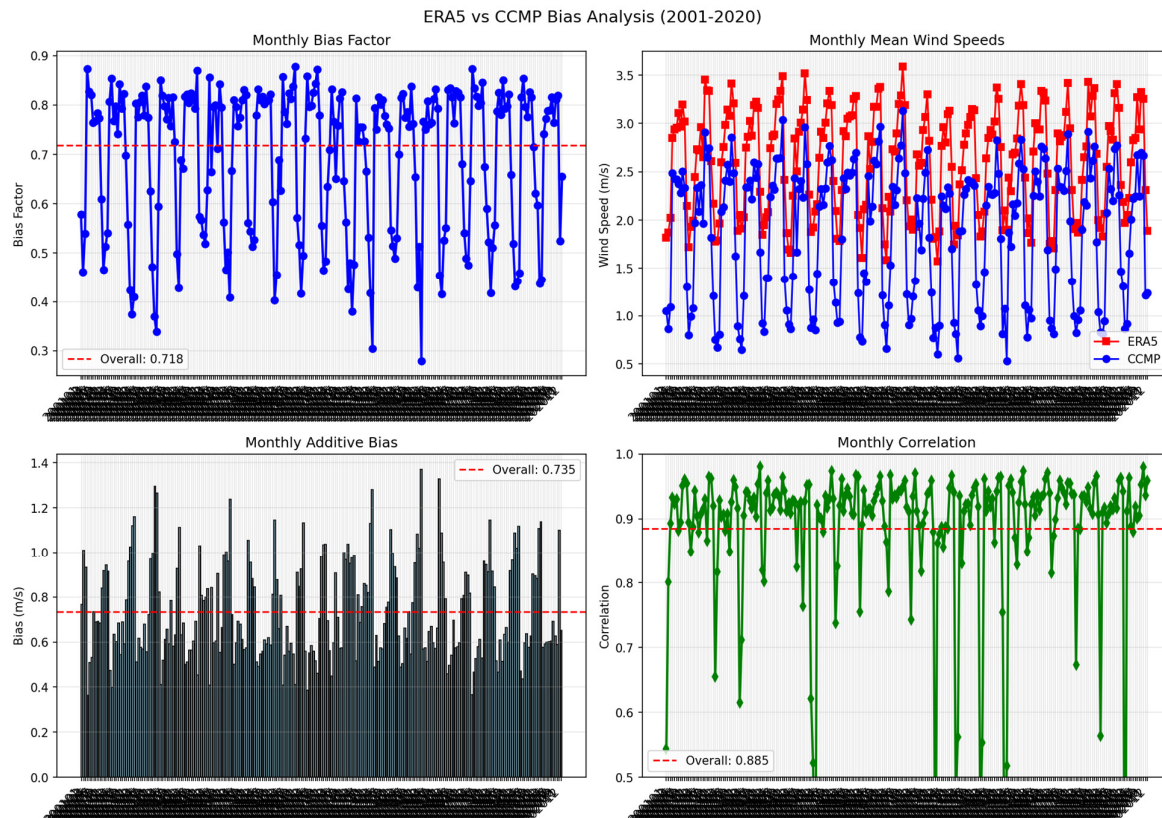
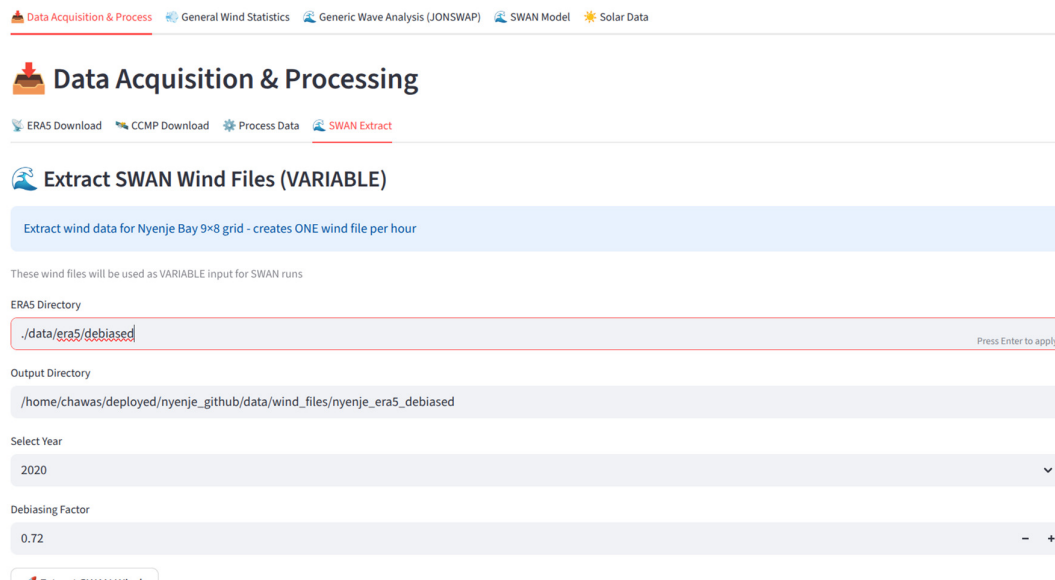


Figure 3-9: ERA-5 Data vs CCMP Bias Analysis

3.10.4 Correlation of Results

The high correlation (0.89) indicates that the spatial and temporal patterns of wind are very well captured by ERA5. This is excellent news: the bias is systematic (a consistent overestimate), not random. Therefore, a simple multiplicative correction is appropriate.

The SWAN extract becomes the next dataset to be processed and be ready as follows:



The screenshot shows a web application interface for data processing. At the top, there is a navigation bar with links: Data Acquisition & Process, General Wind Statistics, Generic Wave Analysis (JONSWAP), SWAN Model, and Solar Data. Below this, the 'Data Acquisition & Processing' section is active, with sub-links: ERA5 Download, CCMP Download, Process Data, and SWAN Extract. The 'Extract SWAN Wind Files (VARIABLE)' section contains a description: 'Extract wind data for Nyenje Bay 9x8 grid - creates ONE wind file per hour'. Below this, a note states: 'These wind files will be used as VARIABLE input for SWAN runs'. The form includes an 'ERAS Directory' field with the value './data/era5/debiased', an 'Output Directory' field with the value '/home/chawas/deployed/nyenje_github/data/wind_files/nyenje_era5_debiased', a 'Select Year' dropdown menu set to '2020', and a 'Debiasing Factor' input field with the value '0.72'. A 'Press Enter to apply' button is located next to the ERAS Directory field.

Figure 3-10: SWAN Extract

3.10.5 General Wind Statistics for Lake Kariba

The second row of table 02 (Methodology Outline) shows five sub-sections under general wind statistics whose images are respectively from 3.1 to 3.5 below

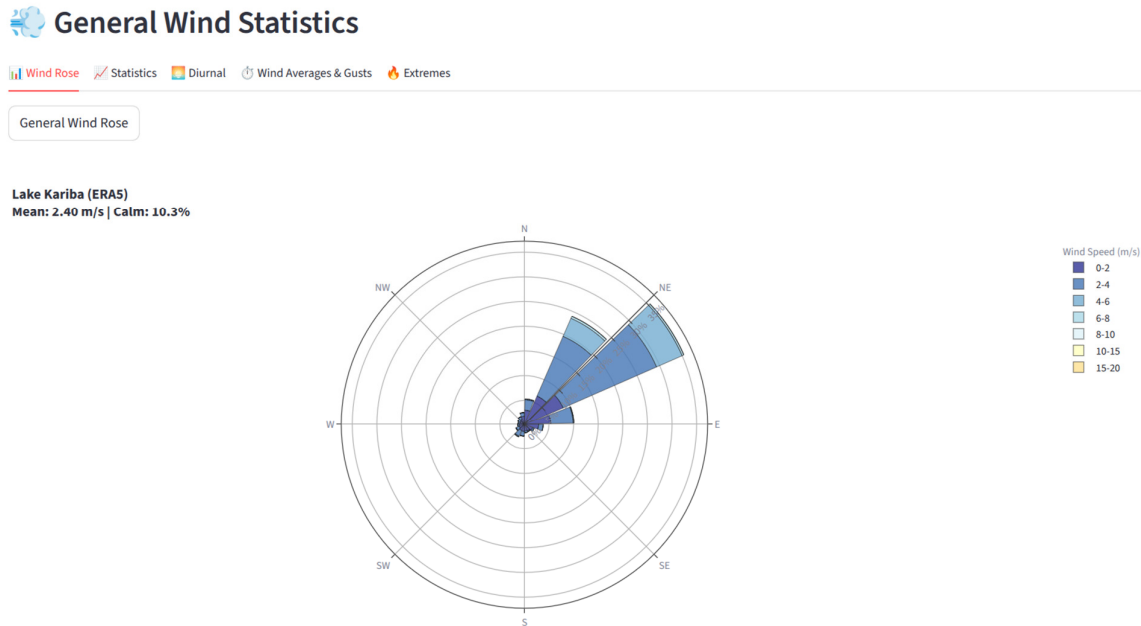


Figure 3-11: Wind Rose - North Easterly - consistent with data from MSD; Padenga & On site

Image 3.1

Key Informatics Wind/Waves Analysis Tool - Nyenje Bay, Kariba

Complete wind, wave, and SWAN wave modeling analysis

Key Concept: Bathymetry is CONSTANT (loaded once). Wind files are VARIABLE (one per hour from ERA5 debiased data).

Data Acquisition & Process General Wind Statistics Generic Wave Analysis (JONSWAP) SWAN Model Solar Data

General Wind Statistics

Wind Rose Statistics Diurnal Wind Averages & Gusts Extremes

Mean Wind Speed	Max Wind Speed	Mean Direction
2.40 m/s	8.90 m/s	51.6°
Median Wind Speed	Year of Max	Cardinal
2.21 m/s	1996	NE

Figure 3-12: Wind Stats in year of highest speed

Image 3.2

Key Informatics Wind/Waves Analysis Tool - Nyenje Bay, Kariba

Complete wind, wave, and SWAN wave modeling analysis

Key Concept: Bathymetry is CONSTANT (loaded once). Wind files are VARIABLE (one per hour from ERA5 debiased data).

Data Acquisition & Process General Wind Statistics Generic Wave Analysis (JONSWAP) SWAN Model Solar Data

General Wind Statistics

Wind Rose Statistics Diurnal Wind Averages & Gusts Extremes

Diurnal Wind Pattern

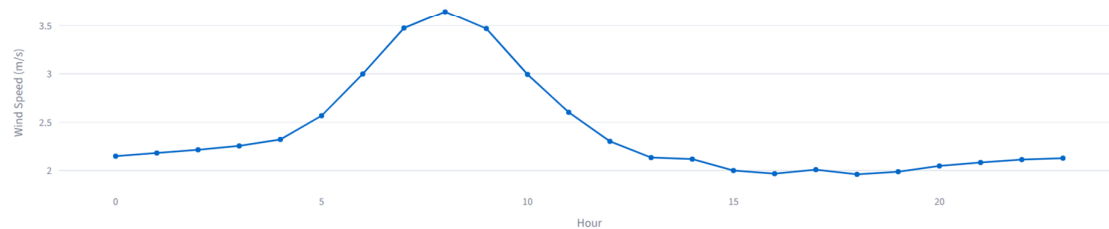


Figure 3-13: Diurnal Results - show generally calm wind conditions over night over a 50-year period

Summary Averages Comparison Wind Rose Time Series Extremes

Wind Averages Comparison

Understanding Wind Averaging Periods & Gust Factors

Why different averaging periods? Wind speed varies significantly over time. Short-term gusts can be much higher than the average wind speed, which is important for:

- **Structural design** - Buildings and structures must withstand peak gusts
- **Wave modeling** - Waves respond to wind over different time scales
- **Emergency planning** - Gusts can cause sudden hazards

	Parameter	Time Period	Gust Factor	Mean (m/s)	Max (m/s)
0	Hourly Mean	60 minutes	1.00 (reference)	1.72	6.39
1	3-Minute Average	3 minutes	1.15	1.72	6.05
2	10-Minute Average	10 minutes	1.00	1.72	5.15
3	3-Second Gust	3 seconds	1.45	2.90	9.27
4	10-Second Gust	10 seconds	1.35	3.05	8.63

Figure 3-14: Wind Averages & Gusts

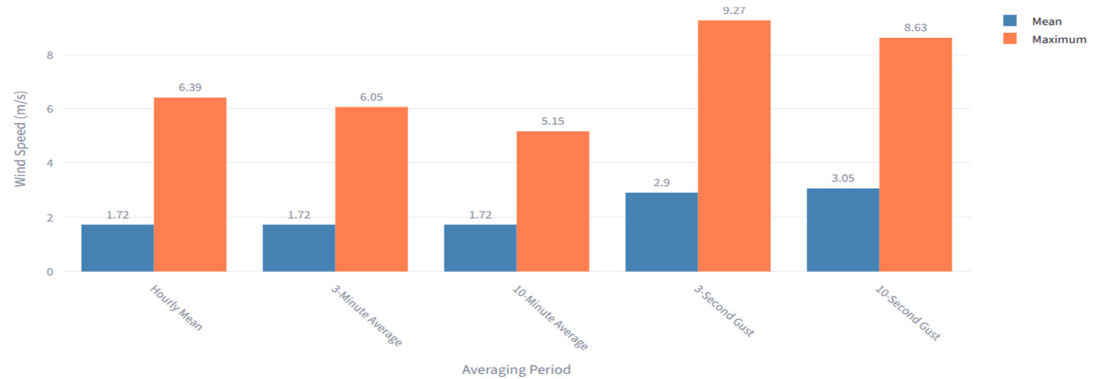
Lake breeze develops 1000-1100hrs and peaks 1300-1500hrs then weakens from 1700-1800hrs.

>>

Deploy

Visual Comparison of Wind Speeds

Wind Speed Comparison by Averaging Period



Understanding the Chart:

- **Blue bars (Mean):** Average wind speed for each averaging period
- **Red bars (Maximum):** Highest recorded wind speed for each period
- The difference between blue and red bars shows the variability
- Larger difference = more gusty conditions

Figure 3-15: Values for Wind speed averages & gusts

Key Concept: Bathymetry is CONSTANT (loaded once). Wind files are VARIABLE (one per hour from ERA5 debiased data).

Data Acquisition & Process General Wind Statistics Generic Wave Analysis (JONSWAP) SWAN Model Solar Data

General Wind Statistics

Wind Rose Statistics Diurnal Wind Averages & Gusts Extremes

Extreme Value Analysis

Annual Maximum Wind Speeds

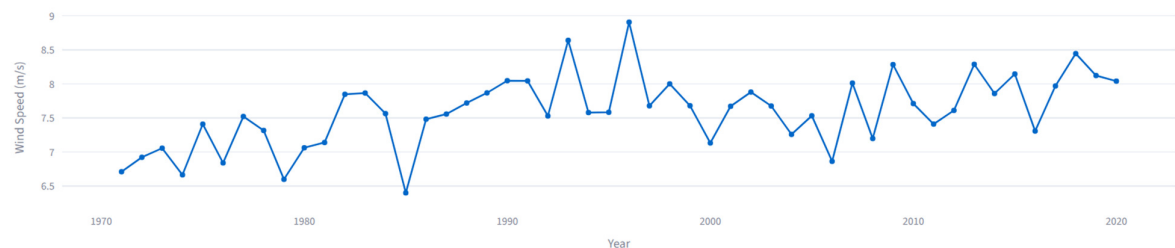


Figure 3-16: Annual Max Wind Speeds

Data Acquisition & Process General Wind Statistics **Generic Wave Analysis (JONSWAP)** SWAN Model

JONSWAP Generic Wave Parameter Analysis

Wave parameters using JONSWAP fetch-limited formula (fetch = 4 km)

Mean Wave Height	Max Wave Height	Sig Wave ($H_{1/3}$)	Mean Wave Period
0.052 m	0.192 m	0.059 m	1.07 s

Wave Height Distribution

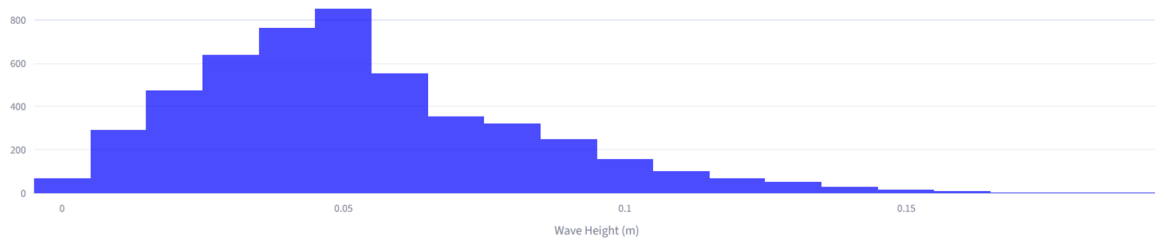


Figure 3-17: JonSwap Typical Results

The JonSwap motivation was a model used to provide some guidance of decent range of results that may be expected for a similar project, and the above set of results was derived ahead of running the SWAN model. There is an interesting correlation with our findings as indicated in the executive summary table.

Furthermore, just some verbal interview and verification with the locals at the nearby Father-girl Fishing Camp revealed that the worst wave heights in the area are typically not above a standard 30cm school ruler as per their observation. Furthermore, given the unique geography of the area with a mountain range on the northern and southern sides plus the My-Trees on the eastern shoreline ahead of another range of mountains, the bay stretches into the main lake westwards, and the winds aren't that harsh naturally.

The return periods shown in figure 3-17 should provide insurable comfort in that the likelihood of waves in excess of 50cm in the project area is not likely considering a 100 years return period.

4 SWAN Model Reporting

Key Informatics Wind/Waves Analysis Tool - Nyenje Bay, Kariba

Complete wind, wave, and SWAN wave modeling analysis

Key Concept: Bathymetry is **CONSTANT** (loaded once). Wind files are **VARIABLE** (one per hour from ERA5 debiased data).

Data Acquisition & Process General Wind Statistics Generic Wave Analysis (JONSWAP) **SWAN Model** Solar Data

SWAN Wave Model Analysis

• **CONSTANT:** Bathymetry (nyenje_bathymetry_9x8.bot)

• **VARIABLE:** Wind files (one per hour from ERA5 debiased data)

Single Analysis Monthly Maxima Yearly Maxima Extreme Value Analysis Find Highest Wave

Single Time Step Wave Analysis

CONSTANT Bathymetry + VARIABLE Wind (select date/time below)

Nyenje Bay Bathymetry (CONSTANT - same for all runs)

Figure 4-1: SWAN Home page

Single Time Step Wave Analysis

CONSTANT Bathymetry + VARIABLE Wind (select date/time below)

Nyenje Bay Bathymetry (CONSTANT - same for all runs)

Nyenje Bay Bathymetry (REAL Data - CONSTANT)

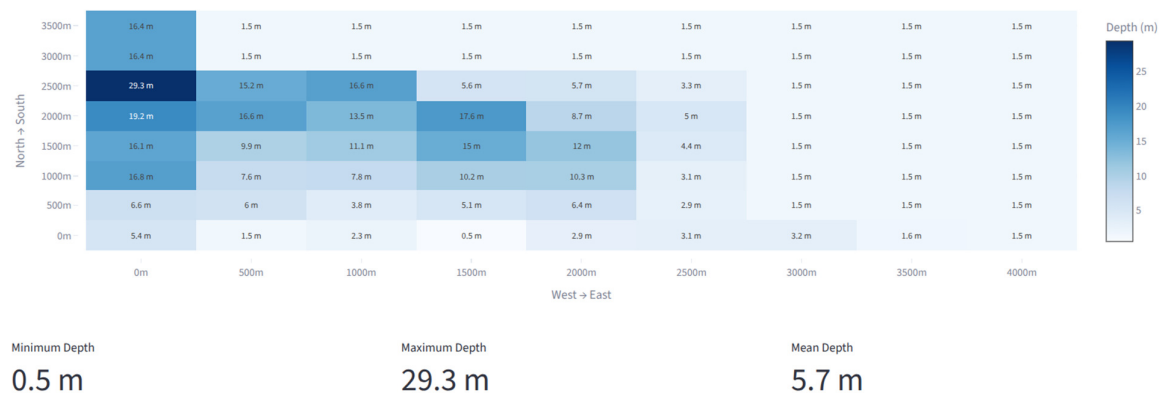


Figure 4-2: Common Bathymetric Information

Select Time Step (VARIABLE Wind)

Year: 2020 Month: 1 Day: 1 Hour: 0

Run SWAN Analysis

Using wind data from: 2020-01-01 00:00:00

Wind Speed

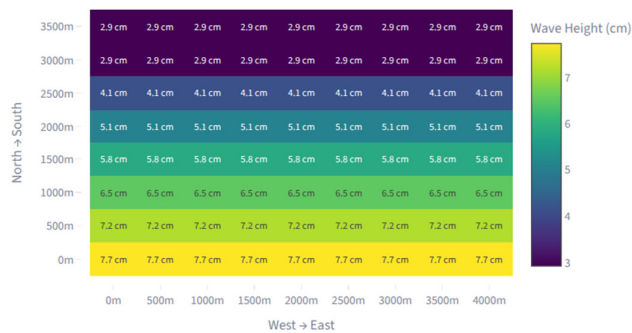
2.58 m/s

Wind Direction

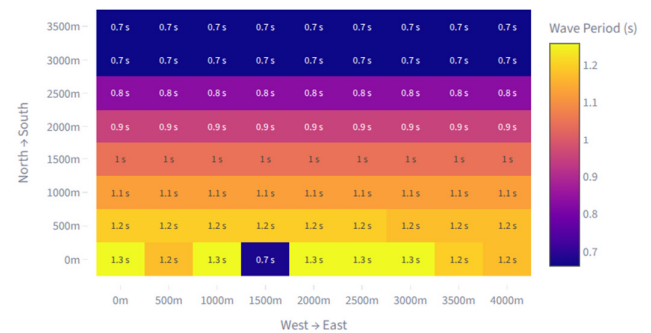
17.0° (NNE)

Figure 4-3: Select Single Time Step

Wave Height Distribution



Wave Period Distribution



Max Wave Height

0.077 m (7.7 cm)

Mean Wave Height

0.053 m (5.3 cm)

Max Wave Period

1.26 s

Mean Wave Period

0.95 s

Figure 4-4: Selected Single Time Wave Height & Period Per Grid

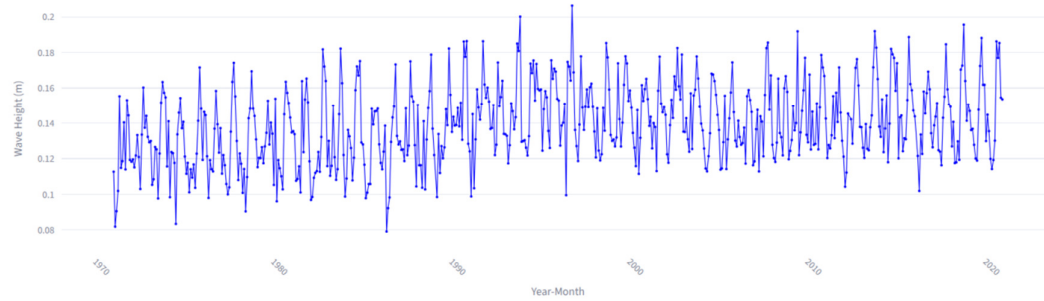
Note that this is a random selection of any date (year, month, day and hour) and the model produces results. This can be demonstrated during the presentation.

Next is the monthly and yearly outcomes over both the period (50 years) and area of interest.

Monthly Maximum Wave Heights

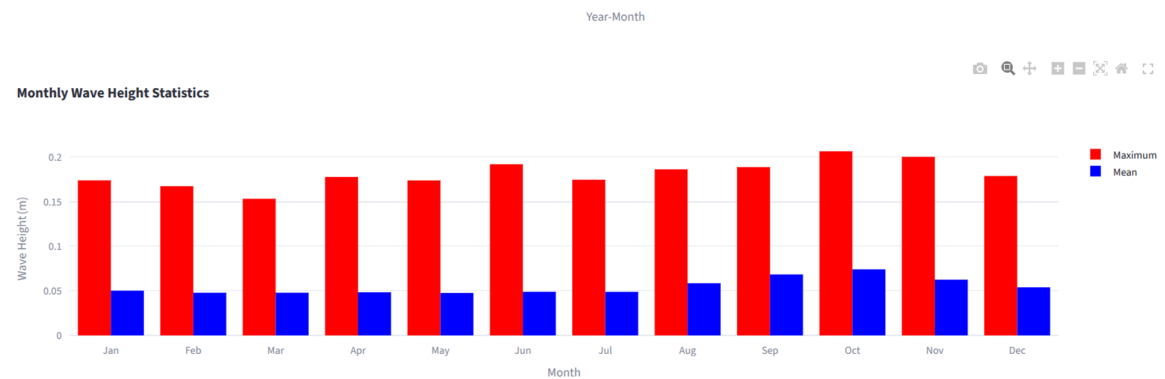
Calculate Monthly Maxima

Monthly Maximum Wave Heights Over Time



Monthly Wave Height Statistics

Figure 4-5: Monthly Max Wave heights



🏆 Overall Monthly Maximum: 0.206 m in 1996-10

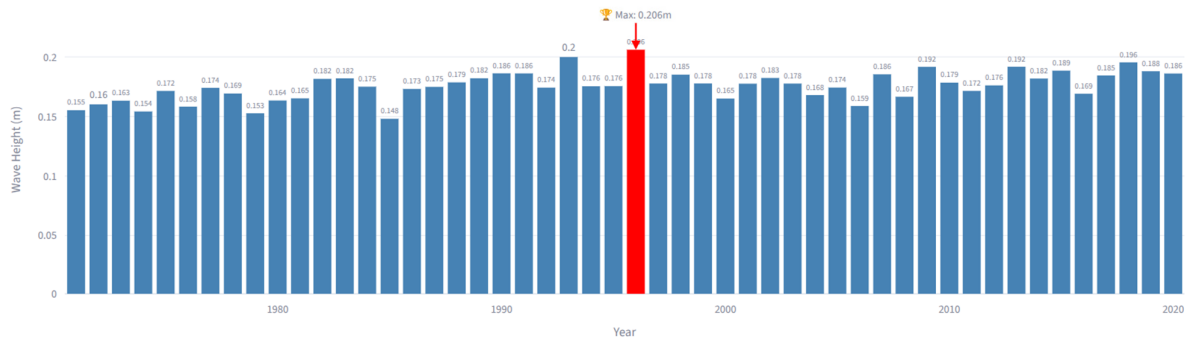
📄 Download Monthly Maxima Data

Figure 4-6: Monthly Wave Height Stats

Yearly Maximum Wave Heights

Calculate Yearly Maxima

Yearly Maximum Wave Heights



Maximum Wave Height

0.206 m

Mean Annual Max

0.176 m

95th Percentile

0.194 m

Year: 1996.0

Figure 4-7: Yearly Max Wave Heights

SWAN Wave Model Analysis

• CONSTANT: Bathymetry (nyenje_bathymetry_9x8.bot)

• VARIABLE: Wind files (one per hour from ERA5 debiased data)

Single Analysis Monthly Maxima Yearly Maxima **Extreme Value Analysis** Find Highest Wave

Extreme Value Analysis (50-100 Year Return Period)

Run Extreme Value Analysis

	Return Period (years)	Wave Height (m)
0	2	0.174
1	5	0.188
2	10	0.197
3	20	0.205
4	25	0.208
5	50	0.217
6	100	0.225

Figure 4-8: Extreme Values And 50-100 Year Return Period

Return Period Wave Heights



Design Parameter	Value	Application
50-Year Return Period Wave	0.217 m	Ultimate Limit State (ULS)
100-Year Return Period Wave	0.225 m	Accidental Limit State (ALS)
Historical Maximum	0.206 m	Recorded extreme
Recommended Design Wave	0.249 m	ULS + 15% safety factor

Download Extreme Value Results

Figure 4-9: Return Periods - Wave Heights

5 Hydrological Summary

Kariba Dam experiences inflows that vary from lows of around 300m³/s to normal annual maximum highs of 3000m³/s-4000m³/s. The highest maximum of 10,000m³/s was experienced during its construction in 1958.

The 6 spillway gates when fully open can discharge 9,500m³/s. This means the maximum flow rate through the lake is about 10,000m³/s whether due to inflowing floods or spillway discharge. These flow rates will induce net downstream currents of about 0.5m/s assuming the water is flowing through the top 1m of the lake and the width of the lake at the point of interest is 20km.

Therefore, it can be concluded the currents induced by the inflows and outflows is insignificant.

6 Structural Design Criteria Memorandum

6.1 Scope and Purpose

This chapter defines the environmental design criteria, characteristic actions, load cases, and load combinations applicable to the structural design of the proposed Floating Solar Photovoltaic (FSPV) facility at Nyenje (Gache-Gache) Bay, Lake Kariba, Zimbabwe.

The provisions herein are derived from the Wind & Waves Study Report (Draft 4, dated 24 April 2026) and are intended to ensure compliance with recognised international and regional standards, while meeting insurer and independent checker requirements.

6.2 Governing Codes and Standards

The structural design shall comply with the following standards, as applicable:

General Structural Design

- EN 1990: Basis of Structural Design
- EN 1991: Actions on Structures
- EN 1993: Design of Steel Structures
- EN 1999: Design of Aluminium Structures
- SANS 10160 (Parts 1–6): Basis of Structural Design and Actions

Marine and Floating Structures

- DNV-ST-N001: Marine Operations and Marine Structures
- DNV-ST-0126: Support Structures for Renewable Energy
- DNV-RP-C205: Environmental Conditions and Environmental Loads

Floating PV Systems

- IEC TS 62738: Design Criteria for Floating PV Systems
- The design shall be based on **Limit State Design (LSD)** principles.

6.3 Design Philosophy

The following limit states shall be considered:

- **Serviceability Limit State (SLS):** Ensuring operability, durability, and user safety without permanent deformation.
- **Ultimate Limit State (ULS):** Ensuring adequate structural resistance under extreme environmental and operational actions.
- **Accidental / Extreme Limit State (ALS):** Ensuring robustness and integrity under low-probability, high-consequence events in line with insurer requirements.

6.4 Site-Specific Environmental Design Parameters

Environmental actions adopted for design are summarised below and are directly derived from the Wind & Waves Study:

i.) Wind

Parameter	Value
Mean wind speed	2.4 m/s
10-minute mean (SLS)	≈ 5.0 m/s
3-second gust (ULS)	15.0 m/s
Accidental design envelope	30.0 m/s

ii.) Waves

Parameter	Value
Mean significant wave height (H_s)	0.18 m
50-year wave height	0.22 m
100-year wave height	0.23 m
Peak wave period (T_p)	1.7–1.8 s

iii.) Lake Currents

Parameter	Value
Maximum design current velocity	0.5 m/s



6.5 Classification of Actions

Permanent Actions (G)

- Self-weight of pontoons / floats
- Structural framing (steel/aluminium)
- PV modules and mounting systems
- Ballast and fixed electrical equipment

Variable Actions (Q)

- Wind actions (W)
- Wave actions (WV)
- Current actions (C)
- Operational and maintenance loads (O)

6.6 Serviceability Limit State (SLS) Load Combinations

In accordance with **EN 1990** and **SANS 10160-1**:

6.6.1 SLS-1: Operational Condition

- G + O

Purpose:

- Float freeboard verification
- Deflection limits
- Functional integrity of PV system



6.6.2 SLS-2: Environmental Condition

- $G + W_{SLS} + WV_{SLS}$

Where:

- W_{SLS} = 10-minute mean wind speed
- WV_{SLS} = mean significant wave height
- Acceptance Criteria:
 - No permanent deformation
 - No ingress of water into floats
 - Acceptable articulation of inter-float connections

6.7 Ultimate Limit State (ULS) Load Combinations – EN / SANS

6.7.1 ULS-W: Wind Governing

- $1.35G + 1.5W$
- Applicability:
 - PV module uplift
 - Frame member strength
 - Edge row stability

6.7.2 ULS-WV: Wave Governing

- $1.35G + 1.5WV$
- Applicability:
 - Pontoon bending
 - Differential float movement
 - Structural connection forces

6.7.3 ULS-COMB: Combined Wind and Wave

- $1.2G + W + WV$

Justification:

- Wind and wave actions are correlated but of low magnitude
- Combination permitted under EN 1990 for correlated variable actions

6.8 Ultimate Limit State – DNV Marine Combinations

For mooring systems, anchors, and global array stability, the following DNV-aligned combinations shall govern:

DNV-ULS

- $1.3G + 1.3E$
- Where:
- $E = W + WV + C$
- Applicability:
- Mooring line design
- Anchor capacity
- Global stability and drift

6.9 Accidental / Extreme Limit State (ALS)

To satisfy insurer and robustness requirements:

- $ALS = G + E_{(Extreme)}$

Where:

- Wind = 30 m/s
- Wave = 0.23 m
- Current = 0.5 m/s



Applicability:

- Progressive failure prevention
- System survivability verification
- Redundancy and robustness checks

6.10 Mooring Design Load Combinations

In accordance with DNV-ST-N001 and IEC TS 62738:

6.10.1 Mooring SLS

- $G + W_{SLS} + C_{SLS}$

6.10.2 Mooring ULS

- $G + W + C$

Acceptance Criteria:

- SLS: Line tension $\leq 40\text{--}50\%$ MBL
- ULS: Line tension $\leq 70\text{--}80\%$ MBL
- No anchor uplift or progressive sliding

6.11 Design Acceptance Criteria

6.11.1 Structural Members

- ULS: Design resistance $\geq$ design action effect
- SLS: Deflection $\leq L/250$ unless otherwise specified

6.11.2 Floats

- Minimum reserve buoyancy $\geq 30\%$
- No full submergence under ULS

6.11.3 Connections

- Fatigue resistance under wave-induced cyclic loading
- No brittle failure modes

6.12 Summary

Based on the adopted site-specific environmental data, the governing load combinations are well within internationally accepted inland FSPV design envelopes. Structural performance is primarily governed by detailing, fatigue, and serviceability, rather than extreme environmental loading.

The load combinations defined in this chapter satisfy the intent and requirements of EN, SANS, DNV, and IEC standards, and are suitable for detailed design, independent checking, and insurance approval

7 Conclusion

This Wind and Waves Study assesses the environmental loading conditions at Nyenje (Gache-Gache) Bay on Lake Kariba to determine the technical suitability and insurability of a proposed large-scale floating solar photovoltaic (FSPV) installation. The assessment is based on a robust 50-year climatological analysis (1971–2020), combining international reanalysis and satellite datasets with available long-term and recent local measurements, and applying industry-standard wave modelling techniques.

Wind conditions were characterised using ERA5 reanalysis data, CCMP satellite observations, and multiple ground-based datasets including Zimbabwe Meteorological Services (Kariba Airport), Padenga Holdings records (≈13 years), and newly installed on-site instrumentation. ERA5 data were conservatively corrected using a bias factor of 0.718 derived from comparison with CCMP winds, ensuring that wind and wave loads are not over-predicted. Wind statistics indicate a low-energy regime dominated by predictable diurnal lake-breeze patterns rather than extreme synoptic events. The long-term mean wind speed is approximately 2.4 m/s, with typical maxima below 9 m/s and short-duration observed gusts around 14–15 m/s. Extreme winds corresponding to 50-year return periods remain well below commonly adopted FSPV design thresholds of approximately 30 m/s.

Wave conditions were assessed using both generic spectral methods (JONSWAP) and detailed site-specific SWAN modelling, incorporating local bathymetry and corrected wind forcing. Results consistently show very mild wave conditions within Nyenje Bay. Mean significant wave heights are approximately 0.18 m, with maximum modelled values around 0.20–0.21 m. The estimated 50-year and 100-year return period significant wave heights are approximately 0.22 m and 0.23 m respectively—an order of magnitude lower than the 1.0–1.5 m wave heights typically used for inland FSPV design and insurance acceptance. Observations and local knowledge corroborate these findings, indicating that wave heights in the bay rarely exceed 0.3 m.

Hydrological conditions were also reviewed. Even under extreme historical lake inflows and spillway discharge scenarios (up to ~10,000 m³/s), resulting currents at the project site are estimated to be no more than approximately 0.5 m/s, with typical operating conditions producing negligible currents. Such current magnitudes do not pose a material risk to floating structures, mooring systems, or electrical infrastructure.

Importantly, Nyenje Bay benefits from significant natural risk mitigation features, including a sheltered bay geometry, limited effective fetch, surrounding mountainous terrain, and vegetated shorelines. These characteristics substantially reduce exposure to extreme wind and wave conditions despite the large size of Lake Kariba.

Based on long-term validated datasets, conservative wind correction, internationally accepted modelling methods, and site-specific SWAN simulations, Nyenje Bay exhibits very low environmental loading from wind, waves, and lake currents. All assessed parameters fall comfortably within, and typically far below, standard design and insurance criteria for inland floating solar PV systems.

It is therefore concluded that the Nyenje Bay site is technically suitable and insurable for the proposed 2 × 500 MW FSPV development from an environmental loading perspective, subject to normal compliance with detailed engineering design standards and routine construction and operational risk controls.
